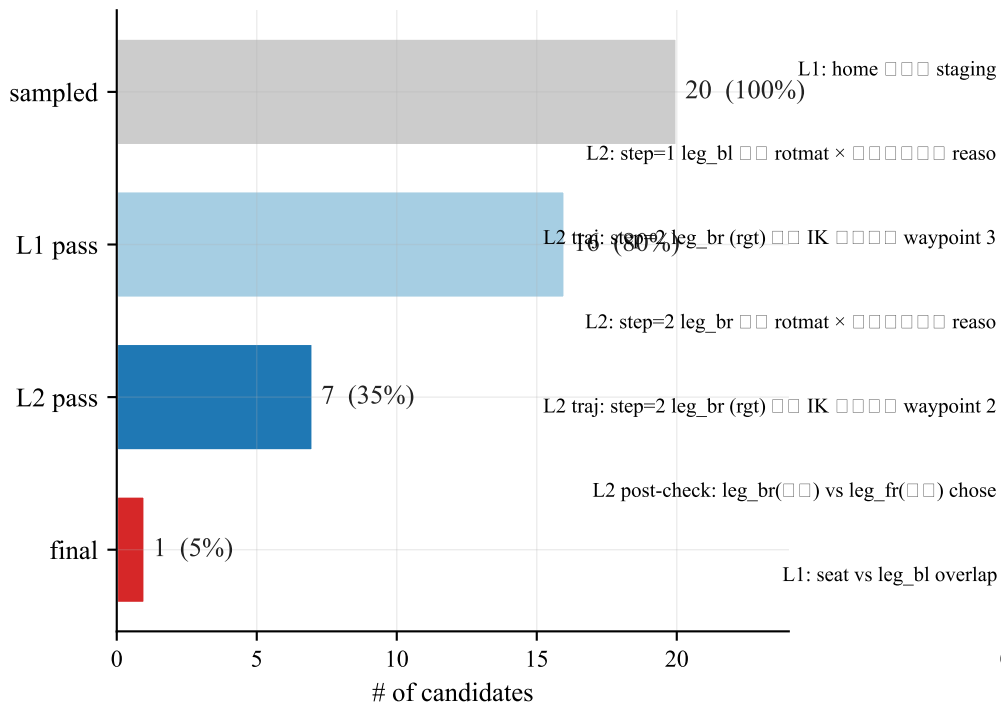


Constraint funnel — the final layout is the unique survivor of a multi-stage hard-constraint filter

Constraint filtering funnel

(every survivor satisfies L1 home-clearance + L2 endpoint IK & collision)



Top failure causes among rejected candidates

